

Multimodality Smartphone App Delivered Therapy for Tinnitus – A Protocol for a Randomised Controlled Trial

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Abstract

Background: Tinnitus is a poorly understood symptom that is common with some patients experiencing severe impacts on their quality of life. The current pathway for patients involves being reviewed by an ear, nose and throat surgeon followed by referral to audiologists who are understandably inundated. Technology is ever advancing and in reducing face to face contact to limit pathogen transmission, smartphone applications may become a solution.

Aims/Objectives: To evaluate the clinical effectiveness of the Oto smartphone app as an alternative to the current standard of care provided on the NHS when looking at tinnitus burden, insomnia severity and quality of life.

Material and Methods: This randomised controlled trial is an open-label, non-inferiority and prospective two arm study. 160 adult participants will be recruited from our secondary care clinics. The patients will be identified based on screening for eligibility criteria. They will undergo randomisation 1:1 to either the Oto application or tinnitus retraining therapy arm. Outcomes measured will use questionnaires that look at tinnitus burden (Tinnitus Functional Index), insomnia severity (Insomnia Severity Index) and quality of life (World Health Organization Quality-of-Life Scale) at baseline, 1 month, 3 months and 6 months post intervention.

Ethics and Dissemination: The trial has already been granted a favourable ethical outcome as per the UK Health Research Authority Bloomsbury Committee (REC:23/LO/0896). Trial results will be made available to the general public through formal publication and dissemination.

Introduction

Tinnitus is a complex and poorly understood heterogeneous subjective disorder causing a “phantom” sound in the ear without an external stimulus. It is extremely common with approximately 7 million in the United Kingdom suffering from it and roughly 10–15% of people experiencing severe impacts on their quality of life. [1]

The current pathway on the national health service (NHS) typically involves being reviewed by an ear, nose and throat (ENT) surgeon or audiologist secondary care and excluding secondary causes for tinnitus such as infective and inflammatory ear disorders, cerebellopontine angle pathology or systemic health problems. Once these are excluded a diagnosis of primary tinnitus is made and the patient is referred for tinnitus therapy or any other management modalities to treat the cause of tinnitus eg hearing aids. According to National Institute of Clinical Excellence (NICE) guidelines, tinnitus therapy includes a combination of tinnitus retraining and sound therapy or cognitive behavioural therapy (CBT). (Fig. 1) [2] However, waiting lists on the NHS are long and this can be expensive to source privately. In a climate where technology is mainstream and integral to our lives, smartphone apps focusing on delivering the above treatment options may be a solution.

A number of apps already exist. Many of these rely on masking therapy as opposed to tinnitus related conditions such as anxiety or insomnia.[3] Internet-based CBT has been shown to be as effective as face to face therapy options when looking at tinnitus burden.[4] The role for telemedicine has been accelerated throughout the COVID-19 pandemic where the need to limit face-to-face contact and reduce virus transmission rates was paramount. This study aims to investigate a UK Medicines and Healthcare Regulatory Agency (MHRA) approved mobile app called Oto which is a novel multimodality app aimed at targeting tinnitus. It combines patient education with CBT, relaxation, mindfulness as well as sound therapy in a customisable package supported by “gamification” elements directed at enhancing user experience. The app offers the user an opportunity to combat their tinnitus at their own pace and choosing the type or combination of therapies that best suits them. The app helps the user reduce their time sacrificed to travelling to a hospital/clinic as well as avoids them having to take time off work. The app can be downloaded from both the Apple App Store and Google Play Store as well as from <https://www.joinoto.com>. The aim of our trial is to evaluate the clinical effectiveness of the Oto smartphone app as an alternative to the current standard of care provided on the NHS when looking at tinnitus burden, insomnia severity and quality of life.

Methods

This will be a single centre, open-label, non-inferiority, prospective, randomised control clinical trial taking place at an NHS centre providing ENT and hearing services with access to tinnitus retraining therapy. The study will look at tinnitus burden, insomnia severity and quality of life in adults with chronic primary subjective tinnitus when using Oto smartphone app compared with NICE approved tinnitus retraining therapy. This will be assessed objectively through questionnaires at the baseline meeting, one month, three months and six months. The estimated duration of the study will be 24 months. The study will be reported in accordance with CONSORT reporting guidelines for randomised controlled trials.[5] Registration with Clinical Trials.gov has been successful (NCT06623123) on 30th September 2024. It is two arm study with the intervention arm being participants being exposed to MHRA approved Oto app (Developed by Oto Health Ltd, App Version 4.3.302, 2021) whilst the control group arm being those that are receiving standard treatment on the NHS but specifically tinnitus retraining therapy (TRT).

Our site will have an ENT and an audiological site lead who will be responsible for registering the study for approval with the relevant Research and Development (R&D) department, advertising the study locally through presentations at departmental meetings, posters in ENT and Audiology outpatient departments, and through Trust websites and email notifications. The ENT and audiological site lead will also be responsible for recruiting patients for the study, checking data accuracy as well as entering study data anonymously on the local trust’s online secure and encrypted database in accordance with data governance policies. The ENT and audiological site lead will have an up-to-date Good Clinical Practice (GCP) certificate and will be responsible for consenting patients.

Adoption of the study onto the National Institute for Health and Care Research (NIHR) Clinical Research Network (CRN) Portfolio will give access to CRN Research Nurses, who can assist in identifying potential

patients and in consenting and recruiting patients into the study. Otology nurse specialists as well as doctors via an associate principal investigator (PI) scheme will also be involved in data collection. Participants recruitment will be from outpatient ENT and hearing services at a single site. The ENT or Audiology Site Lead or NIHR CRN Research Nurse, who will provide eligible patients with a participant information sheet (PIS) describing the study, and will take informed written consent. The ENT surgeon, audiology site lead or NIHR CRN Research Nurse will undertake a daily review of ENT and audiology clinics to identify eligible patients.

Inclusion Criteria:

- Adult patients (male or female) aged over 18 years of age presenting to NHS ENT and hearing services with primary non-pulsatile tinnitus irrespective of hearing.
- Patients must be suffering from tinnitus for a minimum of 3 months
- Patients that have a willingness to use as well as have access to a smartphone device capable of running Oto.
- Willing and able to provide written informed consent.
- Those that would be offered tinnitus retraining therapy (TRT) as management for their primary tinnitus

Exclusion Criteria:

- Patients with secondary tinnitus as confirmed by magnetic resonance imaging (MRI) or other radiological modalities.
- Patients undergoing any other tinnitus therapies concurrently (including hearing aids / implants). Previous tinnitus therapies including CBT in the past are accepted.
- Patients who are not willing to use the app due to not having a suitable phone to download the application or due to own limitations in working Oto.

Randomisation:

The participants that meet the inclusion criteria will be randomised into each treatment arm using an approved internet-based software (www.sealedenvelope.com) [6] by an independent researcher with an allocation ratio of 1:1 and enrolled into either experimental group or control group.

To prevent unequal distribution among groups, participants will be prestratified on factors of age, gender and tinnitus duration at baseline. Block randomization with blocks of two and four will then be utilised to ensure equal sizes groups based on the following categories:

1. Age – under 40, 40–60, over 60 years old
2. Gender – Male, Female

3. Duration of tinnitus – under 1 year, 1–3 years, over 3 years

Due to the nature of the study, blinding is not possible of participants or investigators. Participants will be informed when the intervention would commence but not which type they have been assigned to. Interventions will be applied for the treatment period of 6 months and assessed from the start. Tinnitus functional index (TFI) scores via questionnaire will record the intrusiveness of tinnitus and this will be measured at the start, 1 month, 3 months and 6 months in.

Trial Interventions:

The study will be comparing the Oto smartphone app with tinnitus retraining therapy. A) The **smartphone application** delivers personalized tinnitus therapy combining evidence-based CBT with mindfulness, patient education as well as physical therapy such as stretches and exercises. There are additional sounds available to listen to from the sound library. There will be a structured programme that the participants can work through however they are encouraged to personalise their own therapy.

B) **Tinnitus-retraining therapy** will be offered in the control arm which represents the standard evidence-based intervention on the NHS. [2] Participants enrolled to this arm will undertake tinnitus retraining therapy with a trained audiologist. The sessions will last 30 minutes with the option for further sessions should the patient wish to have more support with their tinnitus to mimic the NHS programme as much as possible. These sessions will encompass a brief audiological history followed by therapy tailored to the patient's needs allowing for patient-centred optimisation of tinnitus burden and secondary comorbidities. This may include relaxation, CBT and advice surrounding sound as well as white noise masking therapy.

In an attempt to standardise intervention arms as much as possible Oto will provide adherence statistics with patients only being included in the final analysis if they have participated for at least 30 minutes in between timepoints throughout the trial pathway.

Primary and Secondary Outcomes:

1. Tinnitus burden via Tinnitus Functional Index questionnaires at 0 months (T_0), 1 month (T_1), 3 months (T_2) and 6 months (T_3). [7]
2. A) Quality of life as calculated via the WHO Quality of Life - BREF questionnaire at the above time intervals.[8]
3. B) Insomnia index as calculated via Insomnia Severity Index questionnaire at the above time intervals.[9]

Statistics and Analysis:

An estimation of sample size was calculated using G*Power version 3.1.6. [10] This was based upon being able to achieve a clinically relevant change between baseline and post intervention using tinnitus functional index as the primary outcome measure. These were calculated using the 13-point difference which concluded that 58 participants were required for each arm of the trial to achieve a two-sided significance level of 0.05 with an effect size of 0.5 and 80% power as per Meikle et al. [7]

An additional 44 participants were recruited to ensure sufficient power during per-protocol analysis to account for possible dropouts. Therefore, 80 participants will be recruited to each arm.

Statistical analysis will be performed with IBM SPSS [11]. An intention-to-treat analysis will be adopted where there are dropouts using the Markov Chain Monte Carlo method of multiple imputation for missing data. 2x4 mixed analysis of variance for repeated measures will be conducted with the within-subject variable of time (T0, T1, T2, T3) and between subject factor of group (Oto app and control) will be carried out with Greenhouse-Geisser correction for non sphericity. This will be followed up by paired samples t-tests to compare within-group differences at each time point followed by independent sampled t-tests to compare outcomes between experimental and control groups. The effect sizes will then be calculated with 95% confidence intervals at postintervention by dividing mean differences by pooled standard deviations. Effect sizes that are below $d = 0.5$ represent small effect sizes whilst those between 0.5 and 0.79 suggest medium effect and those > 0.8 signifying a large impact. [12]

Trial pathway:

Figure 2 below depicts the trial pathway from recruitment to final follow up. **Table 1** below also shows participant timeline: Schedule of enrolment, interventions, and assessments as per SPIRIT 2025 statement. [13]

	Enrollment		Close-out			
TIMEPOINT	-t _i to 0	0 (within 28 days of -t _i)	t ₁ (1 mont h ± 14 days)	t ₂ (3 mont h ± 14 days)	t ₃ (6 mont h ± 14 days)	t ₄ (30 days after t ₃ ± 14 days)
ENROLMENT:						
Eligibility screen	X					
Informed consent		X				
Medical history		X				
Smartphone app education		X				
Randomisation		X				
General welfare check		X	X	X	X	X
INTERVENTION/ COMPARATOR:						
[Smartphone App delivered therapy]		X				
[Conventional Audiology-led Tinnitus Retraining Therapy]		X				
ASSESSMENTS:						
Tinnitus burden via Tinnitus Functional Index questionnaires]		X	X	X	X	
Quality of life as calculated via the WHO Quality of Life - BREF questionnaire]		X	X	X	X	
Insomnia index as calculated via Insomnia		X	X	X	X	

Ethics and Dissemination:

The trial has already been granted a favourable ethical outcome as per the UK Health Research Authority Bloomsbury Committee (REC:23/LO/0896 IRAS ID:307486). Study investigators will adhere to Good Clinical Practice and in line with Research Governance policies to carry out ethically approved research. Patient data will be anonymised. Regular study committee meetings will be held to review progress and highlight any issues that needs to be rectified. The trial results will be disseminated through publication in peer-reviewed journals as well as presentation at relevant conferences. A summary of findings will also be shared with study participants through newsletters. In addition to this, anonymised datasets and

the final study protocol will be made available in an open-access database in keeping with good ethical practice. We propose submitting the main manuscript within 6 months of final data analysis.

Patient and Public Involvement:

Patients and the general public were communicated with in the form of outpatient ENT clinics by authors who expressed dissatisfaction with the current accessibility of tinnitus outpatient management options which involved a suggestive of a telemedicine approach. Patients were not directly involved in the design and conduct of the study, choice of outcome measures or recruitment to the study. Patients/public will be involved in dissemination of results after final analysis through newsletters.

Funding and Competing interests:

Corporate funding has been granted from Oto Health Ltd, 4th Floor, Silverstream House, 45 Fitzroy Street, London, England, W1T 6EB. They will provide licenses for access to the smartphone application in order to run the trial. They will not be involved in conceptualisation, study document formation as well as data collection and analysis. The NIHR has been involved in supporting the study financially.

Discussion

There has been an explosion of smartphone use in medicine to optimise patient centred healthcare. This has been fuelled by the coronavirus pandemic empowering clinicians to find alternatives to limit face to face contact but also aided by a technology focused culture. A lack of randomised controlled trials looking at comparing smartphone applications with conventional evidence based therapies has led to producing the framework for this protocol.

Some key strengths of the protocol are ability to conduct a rigorous and representative study with remote assessments limiting the cost incurred for the investigators while also reducing start up times and increasing access to participants. Dealing with an incredibly common demographic increases the opportunities for recruitment and using a seamless smartphone application with appropriate training will reduce dropout rates. An understandable concern with this study is the ability to standardise the intervention arm with the control arm in terms of usage or adherence of the two. It is important to note that a key advantage of the smartphone application is its accessibility and that should a patient have a relapse in their symptoms they can use the application without the need to be re-referred and await an audiology appointment. In addition, our control arm is based on a treatment offered on the NHS that is evidence-based and whilst the majority of hospitals in the UK offer TRT there may be some subtle differences. Therefore the results of this study may not be transferrable to different demographics across the world.

In summary, this trial was devised to compare whether telemedicine based approaches such as smartphone applications could be used to treat and reduce tinnitus burden, insomnia severity as well as

improve quality of life in adult patients suffering with chronic subjective tinnitus to a degree that is comparable to conventional TRT if not better. This could assist with financial costs and waiting list issues that trouble audiology services. The hope is that studies like this will drive similar trials looking at the role of telemedicine in other aspects of healthcare.

Trial Status

Currently recruiting. Protocol Version 1.3 December 2023. Recruitment began on 4th April 2024 aiming to complete recruitment by August 2026.

Strengths and Limitations

1. **Strong study design:** well structured randomised controlled trial that compares an innovative digital intervention with standardised evidence based treatment such as tinnitus retraining therapy (TRT)
2. **Highly accessible:** use of smartphone based treatments enhances reach to participants, supports telemedicine and reduces healthcare system burden.
3. **Rigorous methodology:** study utilises validated outcome measures, pre-stratified randomisation as well as intention-to-treat analysis to strengthen validity of results.
4. **Lack of blinding:** Open-label design of the study will be a source of potential bias as participants are not blinded to interventions.
5. **Generalisation limits:** Differences in healthcare settings across the world as well as TRT delivery will affect applicability of results outside the UK.

Declarations

1. Clinical Trial Registration details: Registration with Clinical Trials.gov has been successful (NCT06623123) on 30th September 2024
2. Competing interests: there are no competing interests (financial or non-financial) from any of the authors present on this publication.
3. Funding: there were no additional sources of funding for this publication.
4. Ethical approval declaration: this publication has complied with ethical standards; The trial has already been granted a favourable ethical outcome as per the UK Health Research Authority Bloomsbury Committee (REC:23/LO/0896). All data was collected through a hospital audit, was anonymised and disposed of in line with data protection guidance.
5. Consent to participate was in line with Good Clinical Practice standards. Written, informed consent to participate will be obtained from all participants
6. Data availability: Final data sharing will be made available to participants on request and published.
7. Consent for publication has been granted.

8. **Author contributions:** AH designed the work, acquired and analysed data, drafted, revised and approved the manuscript. AH agrees to be accountable for all aspects of the work. AV designed the work, revised and approved the manuscript.
9. Acknowledgements: I would like to thank Mr Ananth Vijendren for his support in finalising the protocol for this trial and his ongoing advice. I would also like to thank Mr Daniele Borsetto and Professor Corazza for their peer reviews and suggestions for the protocol.

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Table 1

Table 1 is available in the Supplementary Files section.

Figures

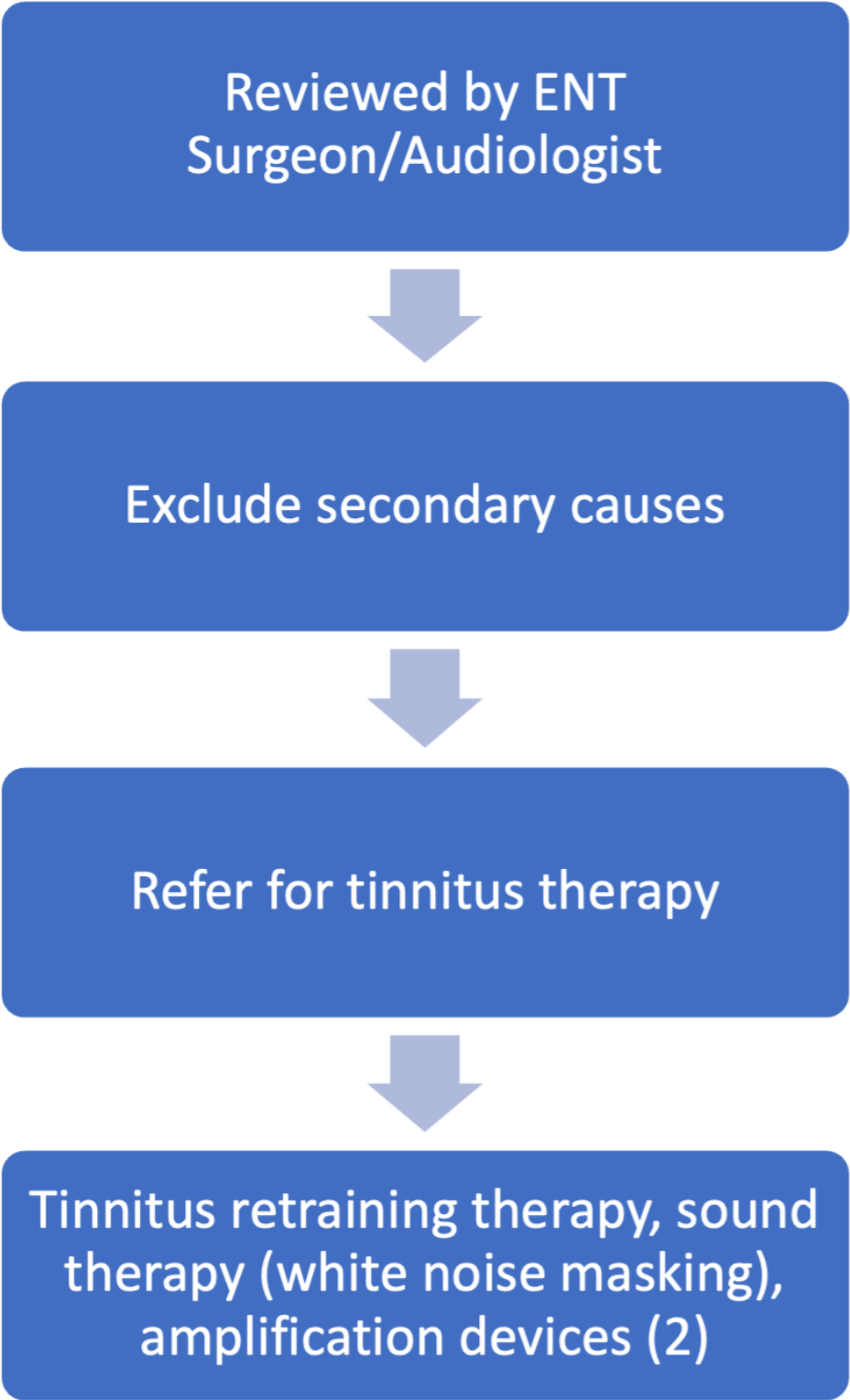


Figure 1

Flow Chart showing the pathway to receiving treatment for Tinnitus in the United Kingdom.

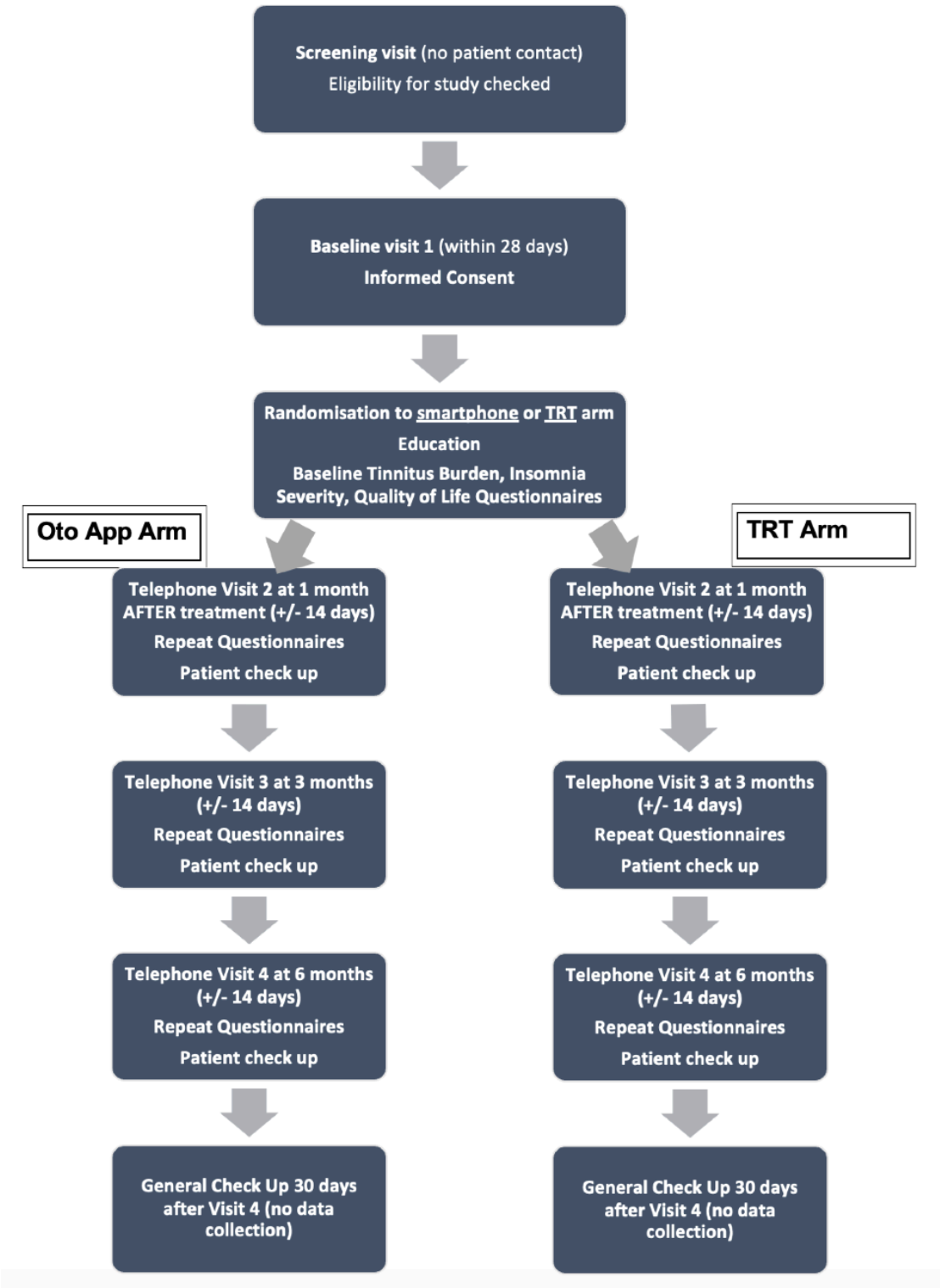


Figure 2

Flow Chart depicting the trial pathway from recruitment to final follow up.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [Table1.docx](#)
- [SPIRIT2025editablechecklist.docx](#)